

DTN A-Score: A Disjunctive-Temporal Generalisation of the Appropriateness Score for Proactive Robotic Assistance

Metric side quest — AURA evaluation

May 5, 2026

Abstract

The Appropriateness Score (A-Score) used to evaluate proactive interventions in the AURA framework assumes a single ground-truth interval per task. Real collaborative tasks admit *alternative* schedules: a step can often be performed in one of several time windows, and choosing one alternative may foreclose another. This note generalises the ground-truth representation and the metric to handle *disjunctive* time specifications, drawing on the Disjunctive Temporal Network (DTN) formalism. We define a schema v2.0 for robot ground-truth files, expose four scoring axes (A_{disj} , A_{act} , A_{cons} , A_{nec}), and validate the metric against a catalogue of thirteen synthetic adversaries on a toy disjunctive scenario and on the existing hand-layup recording. The new metric collapses to the original A-Score on degenerate single-option ground truths, while sharply discriminating strategies that select mutually conflicting options on genuinely disjunctive tasks.

1 Motivation

The A-Score evaluates a proactive intervention track against a robot ground-truth track of intervention events $\{(\text{skill}, t_{\text{start}}, t_{\text{end}})\}$. Each ground-truth event is a single contiguous interval, and a prediction is matched to it by a tolerance window on t_{start} . This works well when the demonstrator’s choice of *when* to do each task is essentially unique, but it fails to represent two important phenomena:

1. **Alternatives.** A task may be legitimately performable in multiple non-overlapping windows. For example, the robot can stage the hardener either before the resin pour begins or in a brief lull partway through it. A model that selects the second window should not be penalised because the demonstrator picked the first.
2. **Multi-phase steps.** A task may consist of two or more intervals that must *both* occur for the task to count as done (e.g. a pre-position and a place; a transport and a return).

Both phenomena are captured by *disjunctive temporal constraints*, which are studied formally in the Disjunctive Temporal Network literature [1]. A constraint of the form $\bigvee_k \bigwedge_j v_{k,j} \in [x_{k,j}, y_{k,j}]$ admits multiple satisfying schedules, each itself a conjunction of windows. Section 2 adapts this formalism to a ground-truth schema, and Section 3 develops a scoring metric that is faithful to it.

2 Disjunctive Ground Truth (Schema v2.0)

2.1 Data structure

A robot ground-truth file declares a list of *interventions*. Each intervention exposes one or more *options*; each option is a non-empty list of *phases*; each phase is a closed time interval $[t_{\text{start}}, t_{\text{end}}]$. Concretely:

- **Phase.** The unit time window. A controllable timepoint must fall in this window for the phase to count as covered.
- **Option.** A conjunction of phases. *All* of an option's phases must be satisfied to claim that option (a multi-step task may demand a pre-condition phase *and* a closure phase).
- **Intervention.** A disjunction of options. *Exactly one* option needs to hold for the intervention to count as done.

The mathematical content is identical to a DTN constraint

$$C_{I_i} : \bigvee_{k=1}^{M_i} \bigwedge_{j=1}^{L_{i,k}} t_{i,k,j} \in [s_{i,k,j}, e_{i,k,j}], \quad (1)$$

where M_i is the number of options for intervention I_i , $L_{i,k}$ the number of phases in option k , and $t_{i,k,j}$ the controllable timepoint to be scheduled inside the corresponding window.

2.2 Backwards compatibility

The schema is a strict generalisation of the existing v1.0 format, in which every intervention carried a single $t_{\text{start}}/t_{\text{end}}$ pair. A v1.0 record is auto-promoted to one option containing one phase ($M_i = L_{i,1} = 1$); the disjunctive axes of the metric defined below trivialise on such inputs.

2.3 Example

The toy scenario used throughout this note is shown in Listing 1 (excerpt from `toy_dtn_three_tasks.robot_gt.json`). Three tasks, each declaring two options; some options are multi-phase. The phase windows overlap deliberately, so that not every combination of options is schedulable on a single agent.

Listing 1: Toy DTN ground truth: three tasks, two options each, with deliberately overlapping phase windows.

```
"interventions": [  
  {"id": "task_1",  
    "options": [  
      {"phases": [{"t_start": 5, "t_end": 10},  
                  {"t_start": 25, "t_end": 30}]},  
      {"phases": [{"t_start": 10, "t_end": 15},  
                  {"t_start": 35, "t_end": 40}]}  
    ],  
  {"id": "task_2",  
    "options": [  
      {"phases": [{"t_start": 0, "t_end": 5},  
                  {"t_start": 10, "t_end": 15}]},  
      {"phases": [{"t_start": 10, "t_end": 15},
```

```

        {"t_start": 25, "t_end": 30}]]}
    ]},
    {"id": "task_3",
     "options": [
       {"phases": [{"t_start": 0, "t_end": 5}]],
       {"phases": [{"t_start": 35, "t_end": 40}]]}
    ]}
  ]

```

Why is this disjunctive scenario interesting? On a single-resource agent (the robot can do at most one task at a time), there is exactly one combination of options that satisfies all three interventions without phase overlap: $(k_1^*, k_2^*, k_3^*) = (1, 1, 2)$, i.e. task 1’s option A, task 2’s option A, task 3’s option B ($\{[5, 10], [25, 30]\}$, $\{[0, 5], [10, 15]\}$, and $\{[35, 40]\}$ respectively). Any other selection forces at least one cross-task phase pair to overlap, making that schedule infeasible. A metric that only checks per-task coverage will fail to penalise such infeasible choices.

3 The DTN A-Score

3.1 Setting

Let the ground truth G comprise interventions $I_1, \dots, I_N$ each satisfying (1). Let the prediction track $P = \{p_1, \dots, p_M\}$ be a set of intervals, each labelled with a skill identifier, agent intervention timestamps and end times. The score combines four axes (all in $[0, 1]$) into a single composite:

$$A_{\text{score}} = w_{\text{disj}} A_{\text{disj}} + w_{\text{act}} A_{\text{act}} + w_{\text{cons}} A_{\text{cons}} + w_{\text{nec}} A_{\text{nec}}, \quad \sum w = 1. \quad (2)$$

Defaults are $(w_{\text{disj}}, w_{\text{act}}, w_{\text{cons}}, w_{\text{nec}}) = (0.4, 0.2, 0.2, 0.2)$.

3.2 Axis 1 — Disjunctive coverage

For each intervention I_i , restrict the prediction track to predictions whose stripped skill matches I_i ’s skill: $P_i = \{p \in P : \text{skill}(p) = \text{skill}(I_i)\}$. For each option k define the option’s mean phase-coverage:

$$\text{cov}(I_i, k) = \frac{1}{L_{i,k}} \sum_{j=1}^{L_{i,k}} \max_{p \in P_i} \text{IoU}(p, [s_{i,k,j}, e_{i,k,j}]). \quad (3)$$

The intervention’s *selected option* is the argmax, $k_i^* = \arg \max_k \text{cov}(I_i, k)$, and the per-task contribution to A_{disj} is $\text{cov}(I_i, k_i^*)$. Aggregating:

$$A_{\text{disj}} = \frac{1}{N} \sum_{i=1}^N \text{cov}(I_i, k_i^*). \quad (4)$$

This axis rewards proposals that fall inside one of the alternative schedules and penalises those that miss every option.

3.3 Axis 2 — Action correctness

For each prediction p , compute its best-IoU phase across *all* intervention/option/phase triples, and declare a tie among phases tied at that maximum. The prediction is *matched* if the maximum is positive and *correct* if any tied phase shares its skill:

$$A_{\text{act}} = \frac{|\{p \in P : \text{matched}(p) \wedge \text{correct}(p)\}|}{|\{p \in P : \text{matched}(p)\}|}. \quad (5)$$

The skill-aware tie-break is essential. Without it, a prediction at $[10, 15]$ (a window that several tasks’ alternative options happen to share) is mis-attributed to the lexicographically-first matching task, spuriously penalising a correctly-aimed oracle.

3.4 Axis 3 — Consistency

The selected options $\{k_i^*\}$ induce a multi-set of *selected phases*. Cross-task overlap among selected phases indicates that the inferred schedule cannot be executed on a single-resource agent:

$$A_{\text{cons}} = 1 - \max_{\substack{i \neq i' \\ 1 \leq a \leq L_{i,k_i^*} \\ 1 \leq b \leq L_{i',k_{i'}^*}}} \text{IoU}\left([s_{i,k_i^*,a}, e_{i,k_i^*,a}], [s_{i',k_{i'}^*,b}, e_{i',k_{i'}^*,b}]\right). \quad (6)$$

A non-conflicting selection scores 1.0; a fully-overlapping pair scores 0.0. This is the axis that captures the user’s ”100% iff non-overlapping options” property in Listing 1.

3.5 Axis 4 — Necessity

Predictions that lie entirely outside every GT phase are spurious. Let $\Phi = \bigcup_{i,k,j} [s_{i,k,j}, e_{i,k,j}]$ be the union of all admissible phase windows. Then

$$A_{\text{nec}} = \frac{\sum_{p \in P} |p \cap \Phi|}{\sum_{p \in P} |p|}. \quad (7)$$

By convention $A_{\text{nec}} = 1$ when $P = \emptyset$ (no false positives can be raised by an empty track; the silent strategy is punished elsewhere through A_{disj}).

3.6 Composite

Combining (2) yields a single number in $[0, 1]$. With the default weights, an oracle scores 1.0, a strategy that selects a mutually conflicting option pair scores at most 0.8 even if it covers every phase with the right skill at the right time, and a silent strategy is bounded above by 0.4 (since $A_{\text{disj}} = 0$ and the consistency axis collapses through tie-breaking).

4 Adversary catalogue

We validate the metric by scoring thirteen synthetic adversaries against the ground-truth track. Each adversary is a deterministic function of the GT, designed to trigger one specific failure mode. The full catalogue is summarised in Table 1.

Adversary	What it does
Oracle (consistent)	Solves the disjunctive CSP for a non-conflicting option assignment; emits one prediction per phase of the chosen options.
Oracle (first option)	Always picks option index 0 for every intervention; may produce conflicting schedules.
Conflict-prone	Greedily picks the option whose phases overlap previously-selected phases the most.
Phase-collapsed	Consistent oracle, but only the <i>first</i> phase of each chosen option is fired.
Half-coverage	Consistent oracle restricted to every other intervention.
Wrong-skill	Right times, but each prediction’s skill label is rotated to the next intervention’s skill.
Near-miss timing	Consistent oracle shifted by +1.5 s.
Far-shifted	Consistent oracle shifted by +8 s.
Premature firer	Consistent oracle shifted by −1.5 s.
Over-extender	Correct phase starts; phase ends padded by +4 s.
One-shot blanket	A single prediction covering the whole task duration with the most-common skill.
Spammer	Rapid-fire short predictions of the most-common skill across the whole task.
Silent	Empty prediction track.

Table 1: Synthetic adversaries used to validate the DTN A-Score.

5 Results

5.1 Toy DTN

Figure 1 shows the GT options stacked above each adversary’s prediction track on the toy three-task DTN. Table 2 reports the four-axis breakdown.

Adversary	A_{disj}	A_{act}	A_{cons}	A_{nec}	A_{score}
Oracle (consistent)	1.00	1.00	1.00	1.00	1.00
Oracle (first option)	1.00	1.00	0.00	1.00	0.80
Conflict-prone	1.00	1.00	0.00	1.00	0.80
Phase-collapsed	0.67	1.00	1.00	1.00	0.87
Half-coverage	0.67	1.00	1.00	1.00	0.87
Wrong-skill	0.67	0.60	0.00	1.00	0.59
Near-miss timing	0.54	1.00	1.00	0.82	0.78
Far-shifted	0.19	1.00	0.00	0.45	0.37
Premature firer	0.57	1.00	1.00	0.87	0.80
Over-extender	0.56	1.00	1.00	0.73	0.77
One-shot blanket	0.04	1.00	0.00	0.56	0.33
Spammer	0.13	0.78	0.00	0.53	0.32
Silent	0.00	1.00	0.00	1.00	0.40

Table 2: DTN A-Score breakdown on the toy DTN (weights $w = (0.40, 0.20, 0.20, 0.20)$).

Discriminations the metric exposes. The table demonstrates that the four axes provide complementary signal:

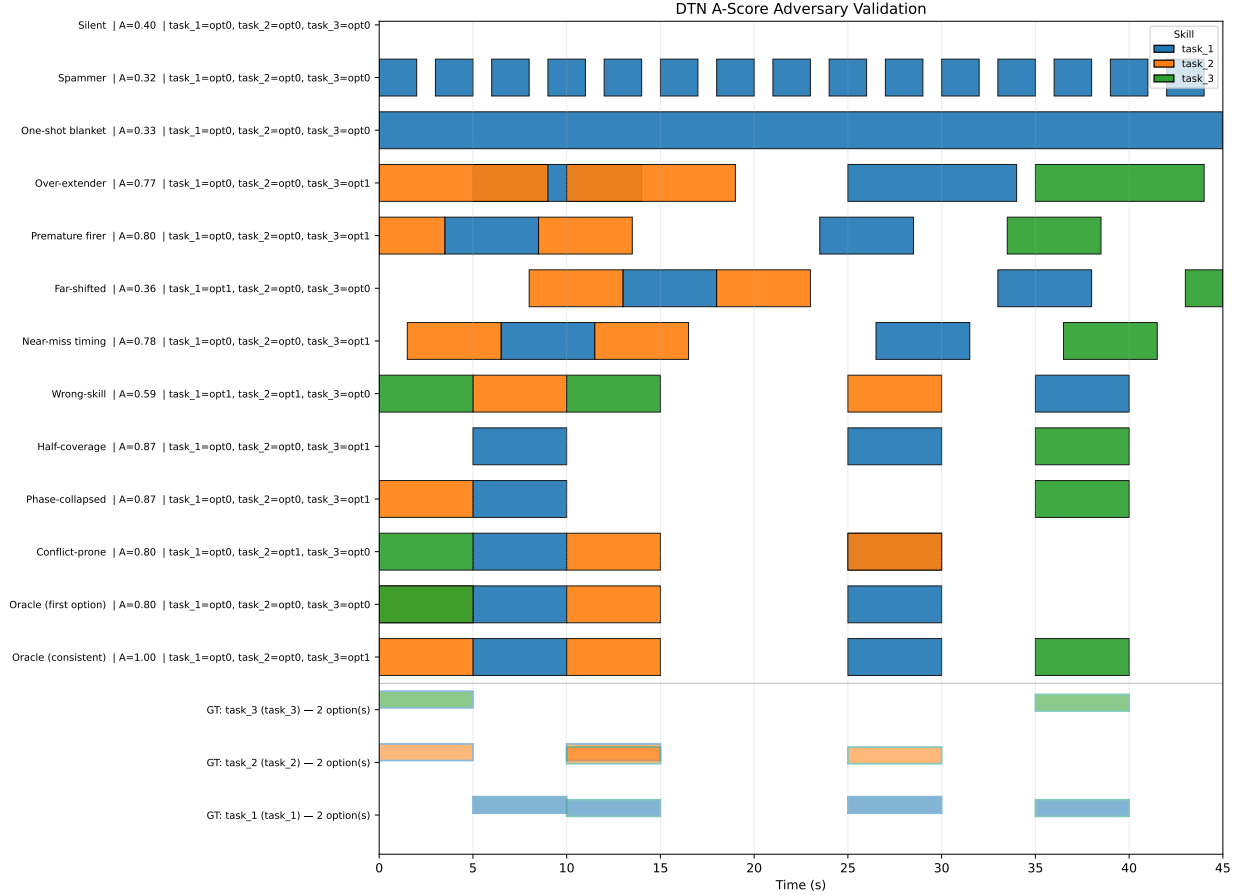


Figure 1: Toy DTN: ground-truth options (top three rows; alternative options drawn with distinct edge colours) above the prediction track of each adversary. The composite A_{score} and the auto-inferred per-task option selection are shown on each row label.

- *Coverage vs. consistency.* Oracle (first option) and Conflict-prone match every GT phase exactly with the correct skill, scoring $A_{\text{disj}} = A_{\text{act}} = 1.0$. They are nonetheless ranked below the consistent oracle because their selected options overlap on a single resource, dropping A_{cons} to zero. This is the user’s failure case: “otherwise, it won’t be able to complete all, and completion will be less than 100%”.
- *Coverage vs. correctness.* Wrong-skill drops A_{act} but also drops A_{disj} and A_{cons} . The auto-selection logic, faced with rotated skills, picks the wrong option for each task and the chosen options happen to conflict.
- *Coverage vs. necessity.* Spammer and one-shot blanket flood the timeline; their A_{nec} correctly drops below 0.6 because most prediction time lies outside any GT phase.

5.2 Hand-layup ground truth (degenerate single-option v2.0)

The existing hand-layup robot ground truth is a degenerate disjunctive file: every intervention has exactly one option containing one phase, so $M_i = L_{i,1} = 1$ for every i . Section 2 guarantees the disjunctive axes collapse on such inputs:

- $A_{\text{cons}} = 1$ for every adversary (no option choice can produce a cross-task overlap that did not exist in the original GT).
- A_{disj} reduces to mean phase-IoU across the single-option, single-phase ground truth, recovering the spirit of the original A_{time} axis.

Figure 2 and Table 3 confirm this: the ranking of adversaries on the hand-layup GT is the qualitatively expected one and matches the original A-Score behaviour, modulo the extra (uninformative) consistency column.

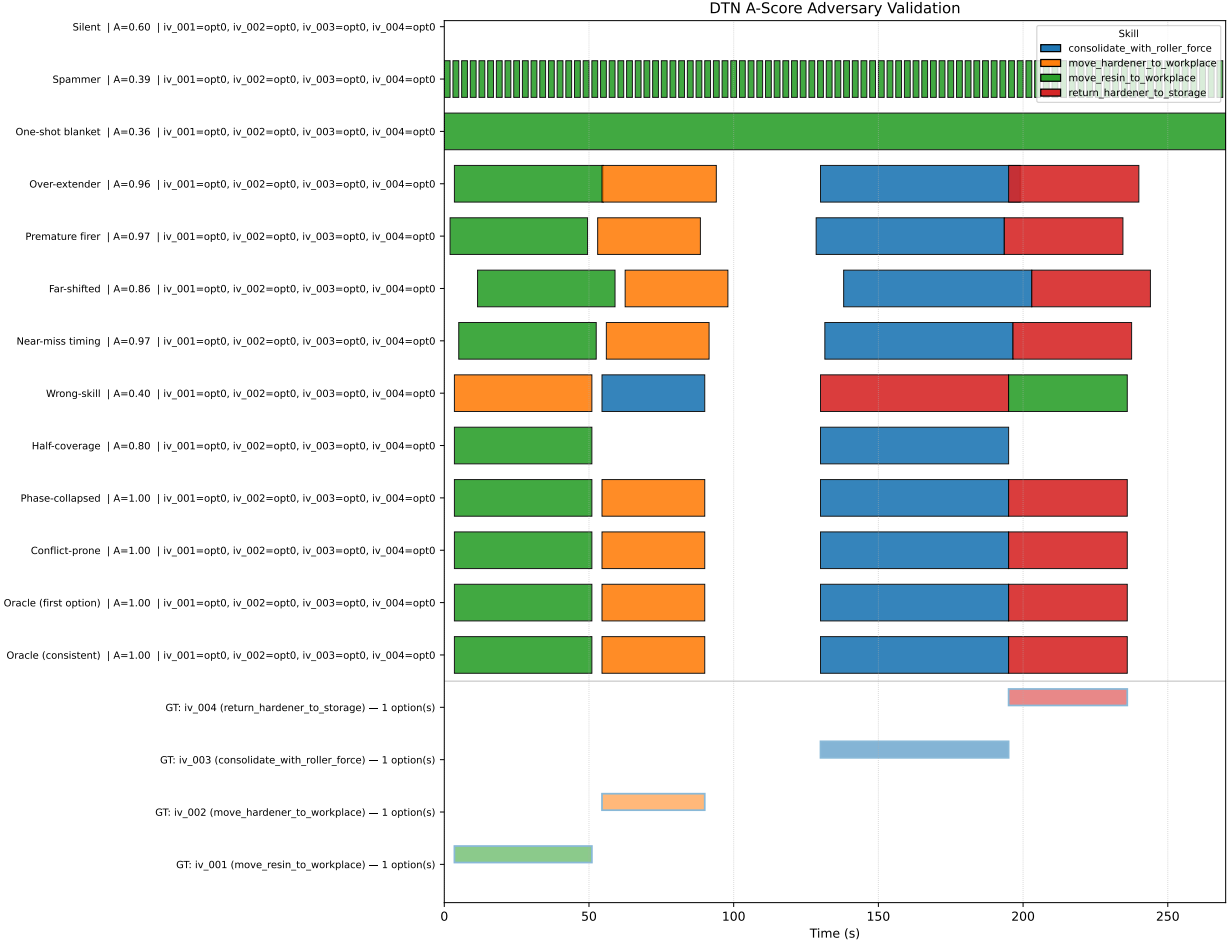


Figure 2: Hand-layup GT (schema v2.0, single option / single phase per intervention): adversary timeline. The disjunctive structure trivialises and the score reduces to mean-IoU coverage with action and necessity penalties; A_{cons} is constant at 1.0 for every adversary.

6 Discussion

6.1 Why A_{cons} rather than a hard feasibility check?

A binary feasibility test (*is the inferred schedule constraint-satisfying or not?*) would discard useful signal: a strategy that overlaps two phases by 1% is qualitatively different from one that

Adversary	A_{disj}	A_{act}	A_{cons}	A_{nec}	A_{score}
Oracle (consistent)	1.00	1.00	1.00	1.00	1.00
Oracle (first option)	1.00	1.00	1.00	1.00	1.00
Conflict-prone	1.00	1.00	1.00	1.00	1.00
Phase-collapsed	1.00	1.00	1.00	1.00	1.00
Half-coverage	0.50	1.00	1.00	1.00	0.80
Wrong-skill	0.00	0.00	1.00	1.00	0.40
Near-miss timing	0.94	1.00	1.00	0.98	0.97
Far-shifted	0.70	1.00	1.00	0.90	0.86
Premature firer	0.94	1.00	1.00	0.98	0.97
Over-extender	0.92	1.00	1.00	0.94	0.96
One-shot blanket	0.04	0.00	1.00	0.70	0.36
Spammer	0.01	0.25	1.00	0.70	0.39
Silent	0.00	1.00	1.00	1.00	0.60

Table 3: DTN A-Score breakdown on the hand-layup GT (degenerate v2.0, single option / single phase per intervention; weights as above).

fully collides on a 5-second window. Using the maximum cross-task pair IoU as the penalty yields a smooth axis, while preserving the property that a feasible selection scores 1.0 and a fully-conflicting selection scores 0.0.

6.2 The silent-strategy quirk

On the toy DTN, the silent strategy scores $A_{\text{cons}} = 0$ rather than the intuitively-expected 1.0. The cause is mechanical: with no predictions to disambiguate the auto-selection, every task ties at coverage zero and the tiebreak picks option index 0 across the board, which happens to overlap on this particular GT. The silent strategy is ultimately punished through $A_{\text{disj}} = 0$, which dominates the composite, so this is not a ranking pathology. A future refinement could mask A_{cons} on tasks with zero coverage; the current behaviour is conservative and defensible (*a strategy that proposes nothing cannot be claimed to be feasible either*).

6.3 Relationship to DTNU controllability

The metric scores a *realised* prediction track, not a strategy in the controllability sense of Osanlou et al. [1]. We borrow the disjunctive constraint formalism (1) but make no use of their tree search or graph neural network heuristic. A natural extension would lift the GT to a full DTNU — modelling human-side actions as uncontrollable timepoints with possible-occurrence intervals — and replace per-trace coverage with a controllability property: *is the policy DC-feasible across every realisation of human timing?* That extension is left for future work.

6.4 Backward compatibility with the existing pipeline

The disjunctive ground-truth loader accepts both schema v1.0 (`t_start/t_end` per intervention) and v2.0 (`options/phases`). v1.0 files auto-promote to one option / one phase, on which the disjunctive metric reduces to the original A-Score’s coverage axis with $A_{\text{cons}} \equiv 1$. The two existing logs (Tea-Making and Hand-Layup) therefore continue to score under the new metric without any change to the recorded ground truth.

7 Reproducibility

The implementation lives entirely in `aura/scripts/eval/metric_side_quest/`:

- `disjunctive_gt.py` — schema v2.0 dataclasses and a loader that auto-promotes v1.0 files.
- `dtn_score.py` — the four scoring axes and the backtracking solver for `find_consistent_selection`, which is used by the consistent-oracle adversary.
- `dtn_adversaries.py` — adversary catalogue and CLI.
- `toy_dtn_three_tasks.robot_gt.json` — the toy GT used in this note.
- `layup_gesture_demo_stationary_with_overlay.robot_gt.v2.json` — a v2.0 copy of the hand-layup GT (degenerate single-option case).

To regenerate the figures and tables in this document:

```
# Toy DTN (Figure 1, Table 2)
python scripts/eval/metric_side_quest/dtn_adversaries.py

# Hand-layup degenerate v2.0 (Figure 2, Table 3)
python scripts/eval/metric_side_quest/dtn_adversaries.py \
  --gt scripts/eval/metric_side_quest/layup_gesture_demo_stationary_with_overlay.
    robot_gt.v2.json \
  --output scripts/eval/metric_side_quest/out/fig_dtn_adversaries_layup.pdf
```

8 Conclusion

We extended the AURA Appropriateness Score to handle disjunctive ground-truth specifications, in which each task admits multiple alternative time-window combinations. The four-axis decomposition (A_{disj} , A_{act} , A_{cons} , A_{nec}) cleanly separates timing coverage, action selection, schedule consistency, and predictive restraint; the new A_{cons} axis is the one that captures the single-resource conflict the original metric could not see. Validation against thirteen synthetic adversaries on a toy three-task DTN and on the existing hand-layup recording shows the metric ranks adversaries as intended and gracefully degrades to the original behaviour on single-option ground truths. The next step is to integrate the loader into the existing evaluation pipeline and revisit any benchmark for which a disjunctive specification is genuinely warranted.

References

- [1] K. Osanlou, J. Frank, A. Bursuc, T. Cazenave, E. Jacopin, C. Guettier, and J. Benton. Solving Disjunctive Temporal Networks with Uncertainty under Restricted Time-Based Controllability using Tree Search and Graph Neural Networks. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2022.